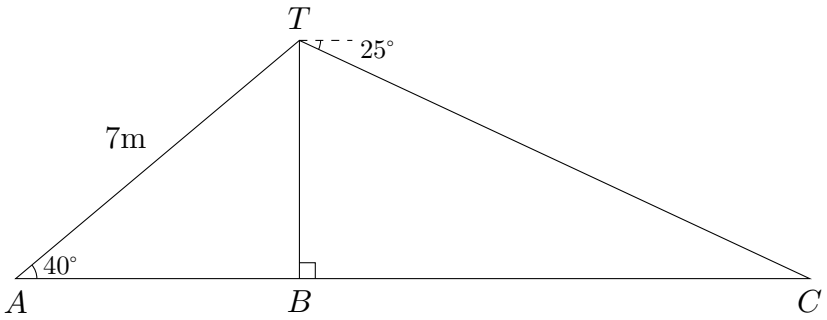
Standard 2 Trigonometry

Question Bank

Question 1

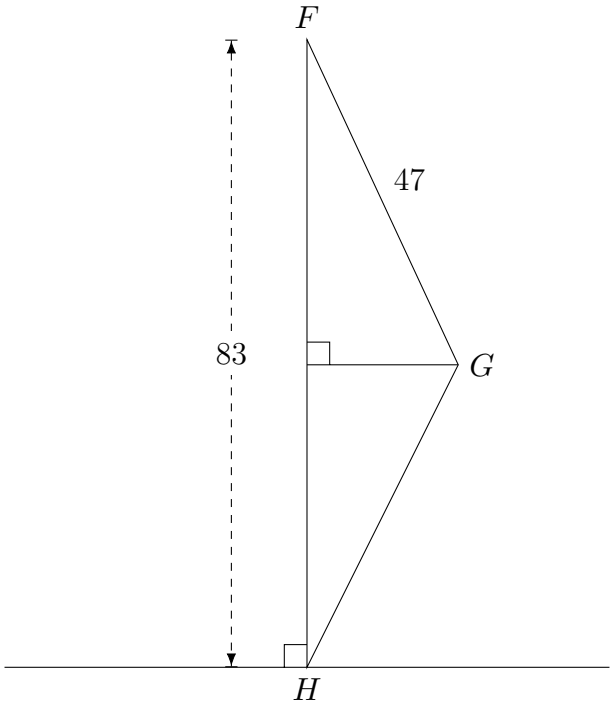
A pole is being held up by two guy wires anchored at A and C. The wire connecting AT is 7m long, and the angle of elevation of T from A is 40°. The angle of depression of C from T is 25°.



How far apart are A and C?

Question 2

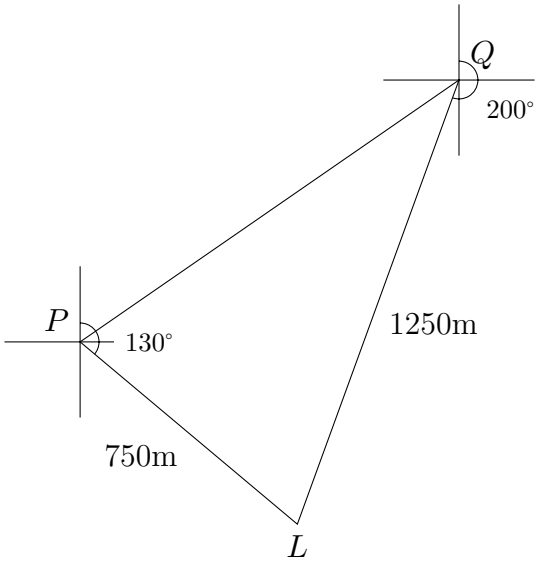
The triangle FGH is shown below. The length of FG is 47 and the length of FH is 83.
The angle of elevation of G from H is 64° and $\angle FGH$ is obtuse.



Find the length of GH correct to two decimal places.

Question 3

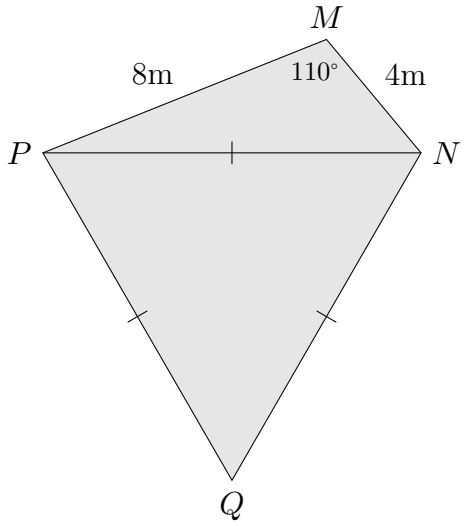
A hiker was at point P and took a measurement of a landmark L that was 750m away on a bearing of 130°. Later when they found themselves lost at Q, they took another measurement of the landmark and found it was on a bearing of 200°and 1250m away.



What bearing has the hiker been walking on, and how far have they gone?

Question 4

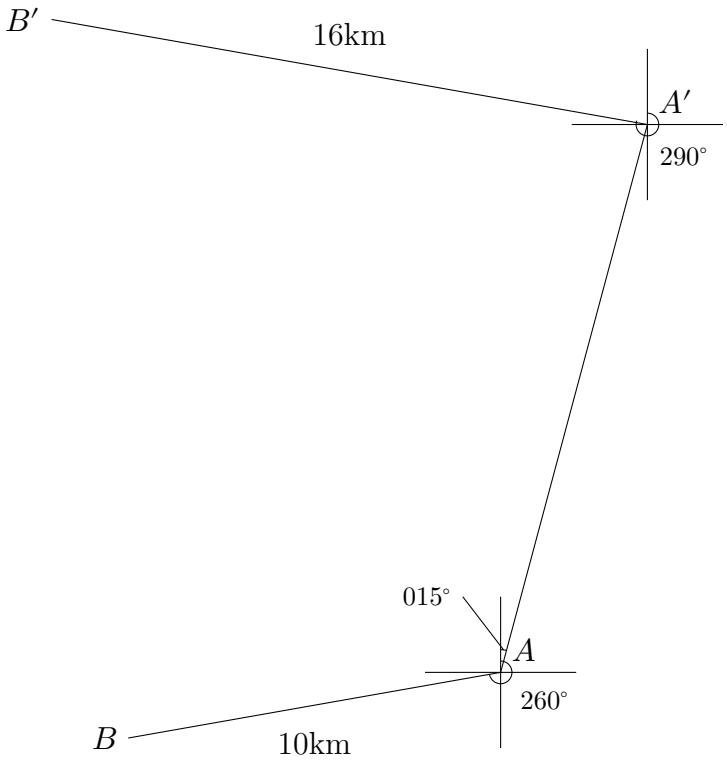
A strange window is constructed from two triangles, PMN and PNQ. The length of PM is 8m and the length of MN is 4m. The $\angle PMN$ is 110°.



The window is to be coated in a protective film that costs \$45/m^2, how much will it cost to coat the window?

Question 5

When Ship 1 is location at position A it determines that Ship 2 is 10km away on a bearing of 260° at point B. Then Ship 1 sails at a speed of 5km/hr on a bearing of 15° for 3 hours, reaching the point A'. Then it made another measurement and found that Ship 2 was 16km away on a bearing of 290° at point B'.



How fast is Ship 2 moving? (Extension question is to find the bearing Ship 2 is travelling on).

Answers

Question 1

Calculate the length of TB,

$$\begin{aligned}\sin(40^\circ) &= \frac{TB}{7}\\ TB &= \sin(40^\circ) \times 7\\ &\approx 4.5\end{aligned}$$

Calculate the length of AB

$$\begin{aligned}AB &= \cos(40^\circ) \times 7\\ &\approx 5.36\end{aligned}$$

Calculate the length of BC using TB, note that $\angle TCB = 25^\circ$,

$$\begin{aligned}\tan(25^\circ) &= \frac{4.5}{BC}\\ BC &= \frac{4.5}{\tan(25^\circ)}\\ &\approx 9.65\end{aligned}$$

Final distance is $AB + BC \approx 5.36 + 9.65 = 15.01m$

Question 2

Since $\angle FHG$ is complementary to 64°,
$\angle FHG = 26^\circ$.
Find $\angle FGH$ using sine rule,

$$\begin{aligned}\frac{\sin(\angle FGH)}{83} &= \frac{\sin(26^\circ)}{47}\\ \sin(\angle FGH) &= \frac{\sin(26^\circ)}{47} \times 83\\ \angle FGH &\approx 50.7^\circ\end{aligned}$$

But since $\angle FGH$ is obtuse,
$\angle FGH \approx 180^\circ - 50.7^\circ = 129.3^\circ$
$\angle GFH = 180^\circ - 129.3^\circ - 26^\circ = 24.7^\circ$

$$\begin{aligned}\frac{GH}{\sin(24.7^\circ)} &= \frac{47}{\sin(26^\circ)}\\ GH &= \frac{47}{\sin(26^\circ)} \times \sin(24.7^\circ)\\ &\approx 44.80\end{aligned}$$

Question 3

The $\angle PLQ = (180 - 130) + (200 - 180) = 70^\circ$.
Use cosine rule to find PQ,

$$\begin{aligned} PQ &= \sqrt{750^2 + 1250^2 - 2 \times 750 \times 1250 \times \cos(70^\circ)} \\ &\approx 1218\text{m} \end{aligned}$$

Use sine or cosine rule to find $\angle QPL$, here I'll use sine rule.

$$\begin{aligned} \sin(\angle QPL) &= \frac{\sin(70^\circ)}{1218} \times 1250 \\ \angle QPL &\approx 74.66^\circ \end{aligned}$$

The bearing of Q from P is $130^\circ - 74.66^\circ = 55.34^\circ$.

Question 4

The area of $\triangle PMN$ is

$$A_{\triangle PMN} = \frac{1}{2} \times 4 \times 8 \times \sin(110^\circ) \approx 15.04\text{m}^2$$

The length of PN

$$\begin{aligned} PN^2 &= 4^2 + 8^2 - 2 \times 4 \times 8 \times \cos(110^\circ) \\ PN &= \sqrt{101.889...} \\ &\approx 10.1\text{m} \end{aligned}$$

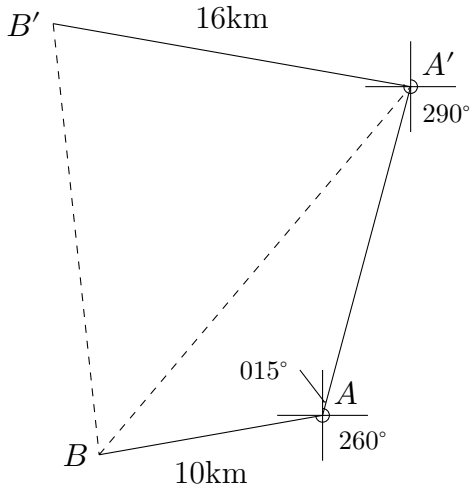
The area of equilateral $\triangle PNQ$

$$A_{\triangle PNQ} = \frac{1}{2} \times 10.1^2 \times \sin(60^\circ) \approx 44.17\text{m}^2$$

The cost is

$$(15.04 + 44.17) \times 45 = \$2664.45$$

Question 5



The length of AA' is $5 \times 3 = 15$.
Find the angle $\angle BAA'$, $360^\circ - 260^\circ + 15^\circ = 115^\circ$.
Use the cosine rule to find the length of $A'B$,

$$\begin{aligned} A'B &= \sqrt{10^2 + 15^2 - 2 \times 10 \times 15 \times \cos(115^\circ)} \\ &\approx 21.26 \end{aligned}$$

Find $\angle BA'A$ using cosine rule

$$\begin{aligned} \angle BA'A &= \cos^{-1}\left(\frac{15^2 + 21.26^2 - 10^2}{2 \times 15 \times 21.26}\right) \\ &\approx 25.22^\circ \end{aligned}$$

Find the angle of $B'A'B$ is $290^\circ - 180^\circ - 15^\circ - 25.22^\circ = 69.78^\circ$
Use the cosine rule to find BB',

$$\begin{aligned} B'B &= \sqrt{16^2 + 21.26^2 - 2 \times 16 \times 21.26 \times \cos(69.78^\circ)} \\ &\approx 21.75 \end{aligned}$$

The speed of Ship 2 is $21.75 \div 3 = 7.25$km/hr.